Project Assignment

REORGANISATION OF A RADIOLOGY DEPARTMENT

Simulation Modelling and Analysis (F000941)
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1. Introduction

In this project, we analyze and improve the performance of a hospital's radiology department using discrete-event simulation. The radiology department is a critical yet resource-intensive part of the hospital, where patients undergo various scans. Delays in this department not only increase patient waiting times but also affect downstream medical processes.

The goal of this project assignment is to identify system bottlenecks and evaluate improvement strategies to reduce the average cycle time of scans and increase the utilization of radiologists.

1.1. System Description

The radiology department under study processes four distinct types of medical scans: X-ray, CT, PET, and MRI. Each scan type follows a specific sequence of workstations (WS1 to WS5), where radiologists are responsible for analyzing the scans. The routing differs per scan type. X-ray scans are routed through WS3, WS1, WS2, and WS5, while CT scans go through WS4, WS1, and WS3. PET scans follow a more extensive route, passing through WS2, WS5, WS1, WS4, and WS3. MRI scans are processed at WS2, WS4, and WS5. These routings are encoded directly into the simulation model via a dictionary structure that links each job type to its corresponding workstation sequence.

We have arrival events and departure events. Scans enter the system through two separate sources. The first source is the radiology department itself, where patients arrive with a mean interarrival time of 0.25 hours. The second source represents referrals from other hospital departments, with a mean interarrival time of 1 hour. In both cases, scan types are assigned based on discrete probability distributions that differ between the two sources. The arrival processes are modeled using exponential interarrival times, which correspond to a Poisson arrival process and are typical for hospital operations.

At each workstation, scans are processed by radiologists. Processing times are assumed to follow a normal distribution, with parameters (mean and variance) that depend on both the scan type and the workstation. If all radiologists at a workstation are busy when a scan arrives, the scan joins a queue that follows a first-in-first-out (FIFO) discipline. This queueing behavior, along with server assignment and job routing, is implemented in a discrete-event simulation (DES) framework. The model progresses by advancing to the next scheduled event, which can be either a scan arrival or a departure from a workstation. We have a departure event when a server finishes processing a job. The job either moves to the next station on its route, or exits the system if done.

Overall, the state of the system at any simulation time *t* is described by following components:

- ***t*** (time)
- ***n*** (number of jobs in the system)
- ***n_ws[i]*** (number of jobs at workstation *i)*
- ***list_scan[i]*** (queue at workstation *i)*
- ***t_a[source]*** (next arrival time from source)
- ***t_d[i][j]*** (next departure time from server *j* at workstation *i*)
- ***current_cust[i][j]*** (customer currently served at workstation *i*, server *j*)
- ***scan_type[job]*** (type of job)
- ***current_station[job]*** (job's step in route)
- ***time_arrival[job]*** (arrival time of the job)
- ***time_departure[job]*** (departure time of the job)

The current staffing of the radiology department includes three radiologists at WS1, two at WS2, four at WS3, three at WS4, and one at WS5. These values are based on the baseline configuration provided in the project assignment. They are fixed in the initial simulation but will be subject to experimentation in later phases of the study.

The simulation itself is developed in Python and is designed to run until 10,000 scans have been processed. During the simulation, key performance indicators such as cycle time (CT) and radiologist utilization (ρ) are recorded.

To evaluate the performance of the system, we use the following objective function:

$$\boldsymbol{Minimize} \qquad \overline{\boldsymbol{CT}} - \mathbf{10}\overline{\boldsymbol{\rho}}$$

- $\overline{CT}$: The average cycle time measured in hours for analyzing the scans.
- $\bar{\rho}$: The average utilization rate that is averaged over all workstations and radiologists.

1.2. Hypotheses

The hospital aimed to explore strategies that involve either adjusting the number of radiologists assigned to WS2 and WS5 or investing in medical software equipment related to WS5, with the goal of minimizing the objective function.

To evaluate these strategies, we formulated the following research hypothesis:

- $\boldsymbol{H_{0_1}}$: The current system design – featuring two servers at WS2, one server at WS5, and the existing WS5 software – performs better than any alternative configuration.

- $\boldsymbol{H_{A_1}}$: There exists at least one alternative configuration that outperforms the current setup.

We also formulated several sub-hypotheses to evaluate specific components of the system design:

Server (radiologists) Allocation at WS2:
- $\boldsymbol{H_{0_2}}$: The current number of servers at WS2 cannot be increased to improve system performance.
- $\boldsymbol{H_{A_2}}$: Adding servers to WS2 improves system performance.

Further analysis investigated the impact of adding more than six servers at WS2:

- $\boldsymbol{H_{0_3}}$: Increasing the number of servers at WS2 keeps improving average utilization.
- $\boldsymbol{H_{A_3}}$: Increasing the number of servers at WS2 beyond a certain point does not improve average utilization.

Server (radiologists) Allocation at WS5:
- $\boldsymbol{H_{0_4}}$: The current number of servers at WS5 cannot be increased to improve system performance.
- $\boldsymbol{H_{A_4}}$: Adding servers to WS5 improves system performance.

Further analysis investigated the impact of adding more than six servers at WS5:

- $\boldsymbol{H_{0_5}}$: Increasing the number of servers at WS5 keeps improving average utilization.
- $\boldsymbol{H_{A_5}}$: Increasing the number of servers at WS5 beyond a certain point does not improve average utilization.

Investment in Medical Software at WS5:

- $\boldsymbol{H_{0}}_6$: No investment in software is needed to improve system performance
- $\boldsymbol{H_{A}}_6$: Investment in software (either upgrade or new) related to WS5 is required to improve performance.

1.3. Overview of Research Test Design

A detailed model of the current system is developed, capturing scan arrival processes, routing through workstations, and scan-specific processing times. Using this model, we evaluate the several redesign scenarios involving staffing adjustments and software upgrades. Our results provide data-driven recommendations to optimize the radiology workflow. We used Python as our programming language.

After analyzing the results of the current system, we began by identifying the appropriate range of radiologist for the screening design. Once this optimal range was established, we generated a set of 75 design configurations. Through an initial screening phase, this set was narrowed down to a representative subset of 6 designs.

Next, we addressed the warm-up period by applying the Welch's method individually to each of the six experimental designs, ensuring the removal of initialization bias. Following this, we implemented the Batch Means Method to analyze system performance. To determine the appropriate batch size for each design, we calculated the covariance of the objective function. To support valid statistical inference, we used 95% confidence intervals based on the batch means, enabling us to assess the precision and reliability of the performance estimates.

To enhance the accuracy of our simulation estimates without increasing the number of replications, we applied variance reduction techniques using antithetic variables.

To gain a deeper understanding of which system parameters had the most significant impact on performance, we conducted a structured screening experiment using a 2^3 full-factorial design. Based on the findings, we proceeded with the comparative design which narrowed our focus to compare specific configurations.

2. Current System Results

In the initial configuration of the radiology department, we simulate the current software setup for WS5 with two radiologists assigned to WS2 and one radiologist at WS5. The simulation reveals strong imbalances in radiologist workload across the system. WS5 clearly emerges as the system bottleneck. This workstation is responsible for processing three scan types (X-ray, PET, and MRI) but is staffed by only one radiologist. As a result, this workstation operates at nearly full capacity, leading to a utilization rate close to 100%. The excessive workload causes substantial queues and delays, which significantly increase the system's overall cycle time.

Meanwhile, other workstations are significantly underutilized. For instance, the average utilization at WS1 is just 17.3%, while at WS2—despite being assigned two radiologists—it drops even lower to 13.2%. This stark contrast illustrates that a large portion of the available radiologist capacity is left idle, highlighting poor server allocation within the system.

The overall system average utilization is 45.27%, meaning more than half of the radiologist capacity remains unused on average. The resulting cycle time is 339.12 hours for 10,000 jobs, but the simulation clearly indicates that this value is unstable, as *Figure 1* shows a continuously increasing trend in the moving average of cycle time.

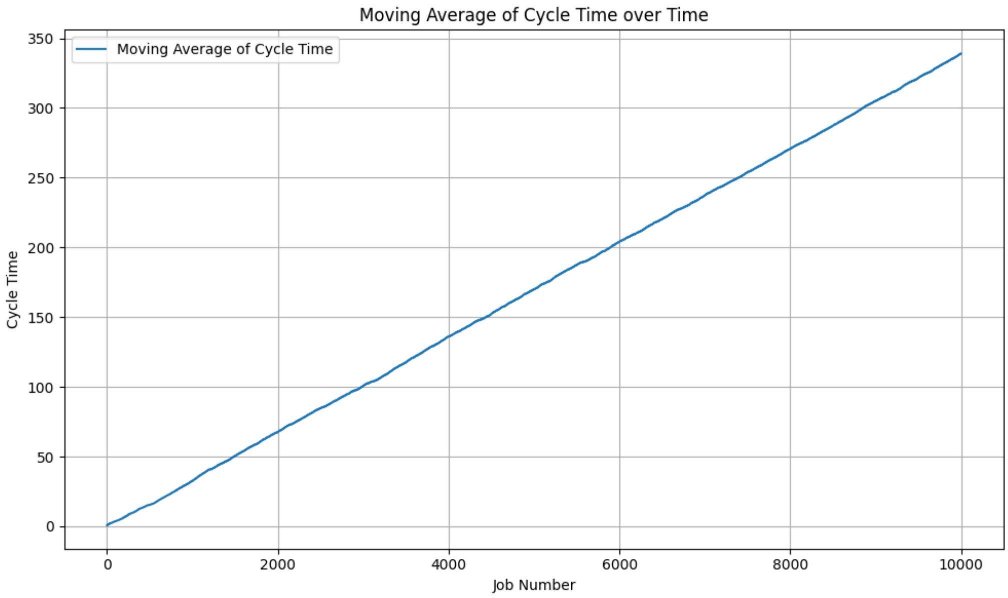


Figure 1: Moving average of cycle time in current situation

Based on the system's objective function, we compute:

$$\text{Objective} = 339.12 - 10 \cdot 0.4527 = 334.59$$

However, it is important to note that this value is not stable. The moving average of cycle time continues to rise throughout the simulation, indicating that the system is fundamentally unstable. As more jobs are processed, the cycle time will continue to grow—further worsening the objective function value. Since we aim to minimize this function, an increasing trend is highly undesirable.

This outcome clearly shows that the current configuration is not viable and urgently needs to be redesigned. In the next sections of this report, we focus on systematically improving the system through a structured redesign, using screening and comparative experiments to guide our optimization process.

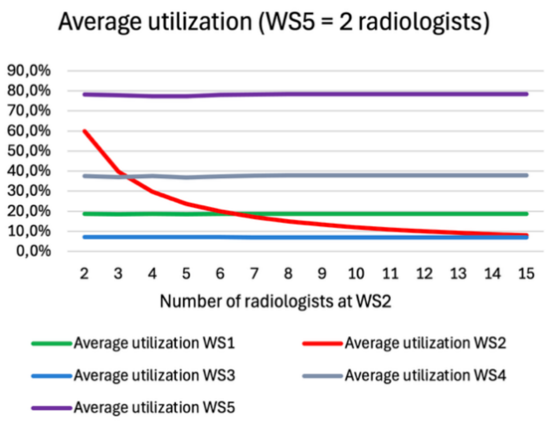
3. Simulation Model Redesign

3.1. Determining the Radiologist Range for Screening Design

In preparation for the screening design, we first noted that two of our input factors — the number of radiologists at WS2 and WS5 — are quantitative variables. For these variables, it was essential to identify a meaningful and realistic range, ensuring that we explore only configurations that could impact system performance. Therefore, we investigated the effect of increasing the number of radiologists at both workstations on utilization. The objective of this analysis was to pinpoint the threshold beyond which additional radiologists no longer contribute to improved performance — in other words, when average utilization at a workstation begins to stagnate — so that we could define an appropriate and efficient range for subsequent experiments.

We conducted this analysis under the current software configuration, as it represents the worst-case scenario in terms of processing times. Since both the upgraded and new software variants require lower processing times, workstation saturation is expected to occur earlier for these alternatives. Thus, by basing our analysis on the current configuration, we ensure sufficient robustness in our screening range.

The graphs below clearly illustrate this phenomenon. *Figure 2 (left)* shows how the average utilization at WS2 decreases as more radiologists are added, while *Figure 2 (right)* presents the same trend for WS5. In both cases, a rapid decline is followed by a horizontal line, indicating that additional radiologists no longer improve performance beyond a certain point.



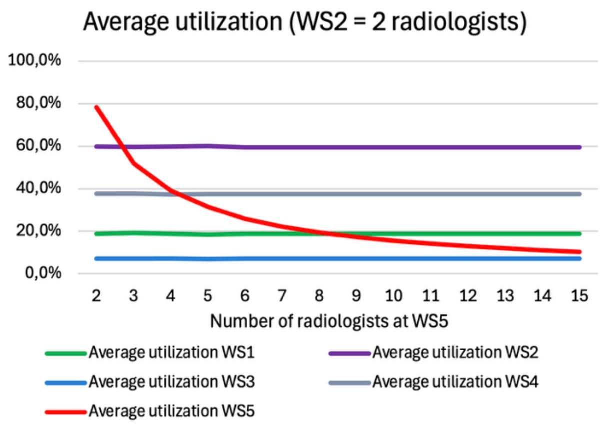


Figure 2: Average utilization for a range of servers at WS2 and WS5

The data used for this analysis can be found in the Excel sheet "***Range number of radiologists***". The results show that:

- For WS2, the average utilization decreases degressively after 4 radiologists, eventually stabilizing near zero.
- For WS5, a similar trend is observed after 4 radiologists.

Importantly, the average utilization of the other workstations remains constant as more radiologists are added to WS2 or WS5. This confirms that the system-wide workload does not increase with more radiologists, as they no longer contribute meaningfully to performance once a certain threshold is crossed. Based on our analysis, we observe that average utilization stagnates after 4 radiologists at both WS2 and WS5. Therefore, sub-hypotheses $\boldsymbol{H_{0_3}}$ and $\boldsymbol{H_{0_5}}$ can be **rejected** – assigning more than 4 radiologists does not yield further in improvement in average utilization.

3.2. Experimental Design

To investigate the performance of the radiology department under different configurations, we constructed a set of 75 experimental designs by systematically varying three key input factors:

- The number of radiologists at workstation WS2 (2 to 5 radiologists)
- The number of radiologists at workstation WS5 (2 to 6 radiologists)
- The system software used at WS5 (current, updated, new)

This factorial design approach enabled a structured exploration of how changes in servers and technology interact to influence system performance. It also allowed us to identify configurations that offer promising trade-offs between resource utilization and throughput.

To enable a robust comparison in the next phase of our analysis, we selected six experimental designs that differ as much as possible in their configuration. This variation ensures that the impact of different factor levels can be meaningfully assessed. In doing so, we ensured that each of the three WS5 software versions is represented by two designs, providing diversity across the software factor.

While maintaining simulation stability, we prioritized variation in factor levels to extract maximum insight from pairwise comparisons. The six selected configurations are listed in *Table 1* and their corresponding Excel sheets are highlighted in red for quick reference.

Experimental Design	Software WS5	# Radiologists WS2	# Radiologists WS5
1	Current	2	3
2	Current	4	5
3	Upgrade	3	3
4	Upgrade	6	2
5	New	4	2
6	New	3	4

Table 1: Choice of different experimental designs

3.3. Welch's Method

Before applying the batch means method, it is essential to determine and remove the initial warm-up period to eliminate the transient effects in the simulation output. To do this, we applied the Welch's method to each of the six experimental designs.

The Welch's method works by smoothing the time series of the running average of the objective function using a moving average with a specified window size, and visually identifying when the output stabilizes (i.e., when the moving averages flatten out). We performed this analysis visually for a range of window sizes to assess the sensitivity of the results.

After testing different window sizes, we selected a window size of 50. This size provides a good balance between sufficient smoothing (to clearly identify trends) and preserving relevant fluctuations (to avoid over-smoothing). A larger window size would obscure the warm-up phase, while a smaller one might exaggerate random noise.

For each of the six selected experimental designs, we plotted the Welch's graph and visually inspected the point at which the moving average of the objective function stabilized, indicating the end of the warm-up phase. While the required warm-up periods varied slightly across the designs, we aimed to maintain consistency and robustness throughout the analysis. Therefore, we applied the maximum observed warm-up period across all designs.

This maximum value was 1195 jobs, identified in the Welch's graph of the design with 2 radiologists at WS2, 3 radiologists at WS5, and the current software installed at WS5. This graph, which is included below as *Figure 3*, clearly shows the point at which the objective function stabilizes, justifying the chosen warm-up threshold. By applying this conservative warm-up period of 1195 jobs to all experimental designs, we ensure that only steady-state behavior is considered in our final performance evaluations.

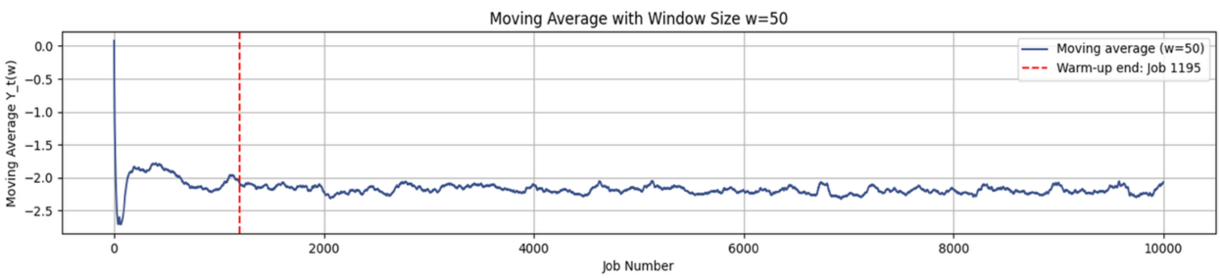


Figure 3: Warm-up period for w = 50

In addition to the warm-up removal, we also accounted for potential end-of-simulation bias, which can occur during the final stage of the simulation when no new jobs arrive and only the remaining jobs are processed. This phase may not accurately reflect steady-state behavior and could skew the performance estimates. As illustrated in the same Welch's graph, *Figure 3* shows this effect toward the end of the run, where the curve starts to move upward. This is undesirable, as it indicates that the objective function value is increasing, which negatively affects performance in our minimization problem.

Rather than manually excluding a fixed number of jobs at the end, this issue is automatically addressed in the batch means method (section 3.4). Specifically, we exclude the very last jobs of the simulation if they are not sufficient to form a complete final batch. These trailing jobs fall outside the regular batch structure and are therefore dropped from the analysis to ensure that only representative and statistically valid steady-state data is retained.

3.4. Batch Means Method

To conduct statistically valid analysis on our simulation results, we applied the batch means method. In discrete-event simulations, consecutive output observations are typically autocorrelated, violating the independence assumption required for standard confidence interval estimation. This autocorrelation motivates the use of the batch means as a variance estimation technique.

Instead of treating each individual observation as an independent data point, we divide the output from a long simulation run into several non-overlapping batches of consecutive observations. Each batch yields an average value, and these batch means are treated as approximately independent for the purposes of statistical inference. We applied this method to all six chosen experimental designs we wanted to further investigate. The designs can be found in *table 1*.

To determine a suitable batch size, we examined the covariance structure of the objective function. Specifically, we calculated the lag L for which the absolute value of the covariance dropped below 0.01, using the following criterion:

$$\textbf{Find L such that } |\boldsymbol{Cov}(\boldsymbol{X_i}, \boldsymbol{X_{i+L}})| \approx \mathbf{0}$$

indicating statistical independence at that distance. To ensure batch independence, we conservatively set the batch size to $5 \times L$.

Based on this method, we found a suitable value of L = 100, resulting in a batch size of 500. Consequently, each simulation run was divided into 17 batches. This choice ensures that the maximum lag-based covariance remains well below the 0.01 threshold, meeting the independence requirement.

Importantly, this batch size was chosen based on the worst-performing design in terms of statistical stability—Upgrade 3–3—which we used as a robustness benchmark. The covariance decay for this scenario is illustrated in *Figure 4*, clearly showing that the covariance falls below 0.01 around lag 100. This conservative choice guarantees reliable statistical inference across all experimental designs.

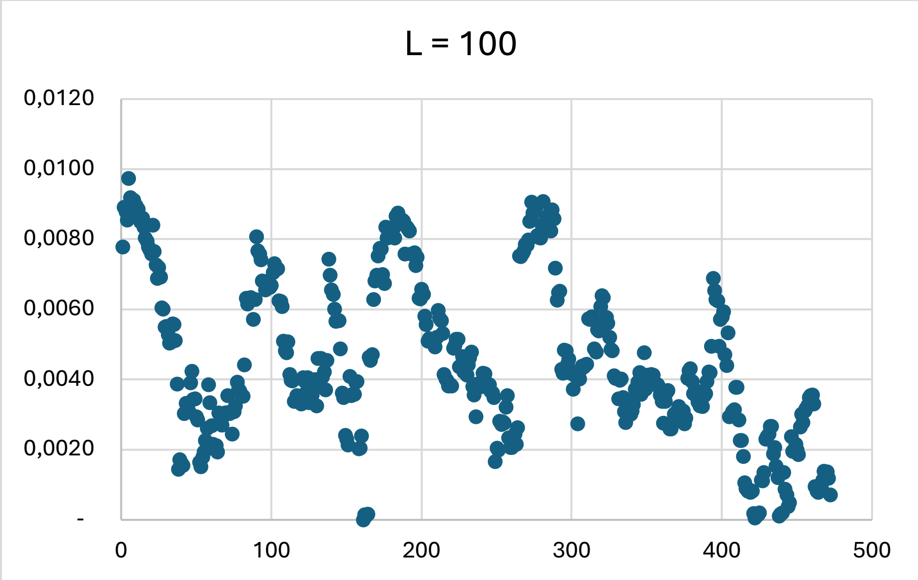


Figure 4: Covariance between observations of experimental design 'Upgrade 3-3' and L = 100

Using these batch means, we computed 95% confidence intervals (section 3.6), coefficients of variation, and relative precision for each experimental design. We further validated the independence of the batches using the Ljung-Box test and lag-1 autocorrelation. If signs of autocorrelation persisted, we recommended increasing the batch size further.

This method allows us to draw valid statistical conclusions from a single long simulation run, ensuring that the reported performance metrics reflect steady-state system behavior rather than transient (warm-up) effects.

3.5. Variance Reduction using Antithetic Variables

To obtain more accurate simulation estimates, it is often useful to reduce the variance of the output. One widely used technique for this purpose is the method of antithetic variables. This approach improves the precision of Monte Carlo estimates by introducing negative correlation between paired simulation outputs. As a result, random fluctuations tend to cancel out when averaging, leading to a lower overall variance without the need for additional simulation runs.

The concept is straightforward: from a single random number $U \sim \mathcal{U}(0,1)$, we generate a pair of inputs —U and its antithetic counterpart $1 - U$. These are then fed through the simulation model, producing two dependent outputs, denoted X and X'. If the simulation output function is monotonic with respect to the random input, X and X'will typically be negatively correlated, which is the key property for variance reduction.

The variance of the average of the two outputs is given by the following expression:

$$Var\left(\frac{X + X'}{2}\right) = \frac{1}{4}[Var(X) + Var(X') + 2\,Cov(X, X')]$$

When the covariance $Cov(X, X')$ is negative, this expression is smaller than the variance of either X or X' alone, thereby improving the precision of the estimate.

In our simulation study, we applied this technique by adapting the exponential distribution sampling method. Specifically, we paired each random number U with its complement 1− U and generated two service time samples accordingly. This was implemented directly in the code (see python file '*Distributions_antitethic*').

We extended this analysis using the batch means method with a batch size of 500, resulting in 17 batches. The antithetic variables method was then applied consistently within this batch structure by simulating each batch twice — once with the original random numbers and once with their antithetic counterparts. This paired design ensures negative correlation between the two paths, thereby reducing variance in the final performance estimates (all calculations can be found in the accompanying *red Excel sheets*).

3.6. Confidence Interval Estimation

Since we are estimating the population mean based on a relatively small sample size (n = 17 batches), we used the t-distribution rather than the normal distribution. This adjustment accounts for the additional uncertainty in estimating the population standard deviation from a small sample. The general formula for the confidence interval of the mean is given by:

$$\bar{x} \pm t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}$$

Converting this into an explicit form for interval notation yields:

$$\mathrm{CI} = \left[\bar{x} \pm t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}\ ;\ \bar{x} \pm t_{\alpha/2,n-1} \cdot \frac{s}{\sqrt{n}}\right]$$

For the experimental design using the current software configuration with 2 radiologists at WS2 and 3 radiologists at WS5, we derived the following summary statistics based on the batch means method (17 batches of size 500):

- Mean objective function value: $\bar{x} = -2.16$
- Sample standard deviation: $s = 0.03$
- Number of batches: $n = 17$
- Corresponding t-value for a 95% confidence level and $n - 1 = 16$ degrees of freedom: $t_{0.025,16} \approx 2.12$

Substituting these values into the confidence interval formula:

$$\mathrm{CI} = \left[-2.16 - 2.12 \cdot \frac{0.03}{\sqrt{17}}\ ;\ -2.16 + 2.12 \cdot \frac{0.03}{\sqrt{17}}\right] = [-2.17597\ ;\ -2.14690]$$

This narrow confidence interval reflects the low variability across batch means, indicating a high degree of precision in the estimated objective function for this system configuration.

4. Experimental Design

4.1. Screening Design

To gain a deeper understanding of which system parameters most significantly affect performance, we conducted a structured screening experiment. The primary objective of this screening phase was to identify the main drivers and key interactions that influence the system's objective function, enabling us to focus subsequent analyses on the most impactful factors. Rather than testing all possible configurations across a broad design space, a screening design allows for a targeted and efficient exploration of the most relevant input variables.

We opted for a 2^3 factorial design, varying three key input factors, each at two levels:

- **Factor 1:** Software at WS5 – Current vs. Upgrade/New
- **Factor 2:** Number of radiologists at WS2 – 2 vs. 4
- **Factor 3:** Number of radiologists at WS5 – 2 vs. 4

For factor 2 and factor 3, the upper and lower bounds were defined based on the workstation utilization analysis discussed in 3.1. That analysis demonstrated that adding more than 4 radiologists at WS2 or more than 4 at WS5 did not further improve system performance, justifying the selected ranges.

A 2^3 full-factorial design was chosen because it offers a balanced and statistically powerful approach to estimate main effects and two-way interaction effects without requiring a prohibitively large number of runs.

Each combination of factor levels was run 17 times (representing the 17 batches in the simulation determined earlier in the batch means method). This replication was necessary to ensure the statistical reliability of the results. Using the batch means method, we computed the average objective function value, standard deviation, and confidence intervals for each combination. These outputs were then used to estimate the main effects and test their statistical significance using confidence intervals and a t-distribution (with 16 degrees of freedom).

4.1.1. Screening Design 1

In this first screening design, we evaluated the impact of three key input factors on the performance of the radiology system:

- **Software at WS5** (Factor 1): Current vs. Upgrade
- **Number of radiologists at WS2** (Factor 2): 2 vs. 4
- **Number of radiologists at WS5** (Factor 3): 2 vs. 4

Each factor was tested at two levels (low and high), resulting in a 2^3 full-factorial design with eight experimental configurations. *Table 2* shows the design matrix and corresponding response labels.

Design	Factor 1 (Software WS5)	Factor 2 (Radiologists WS2)	Factor 3 (Radiologists WS2)	Response
1	Current (-)	2 (-)	2 (-)	R1
2	Upgrade (+)	2 (-)	2 (-)	R2
3	Current (-)	4 (+)	2 (-)	R3
4	Upgrade(+)	4 (+)	2 (-)	R4
5	Current (-)	2 (-)	4 (+)	R5
6	Upgrade (+)	2(-)	4 (+)	R6
7	Current (-)	4 (+)	4 (+)	R7
8	Upgrade (+)	4 (+)	4 (+)	R8

Table 2: Screening design 1: Factor settings and simulation responses

The full results of this screening design, including the main effects, standard deviations, confidence intervals, and significance evaluations, can be found in the Excel sheet **"Screening 1"**.

We assess statistical significance based on whether the 95% confidence interval includes zero. If zero is not contained within the interval, the effect is deemed statistically significant. In this case, all three main effects (F1, F2, and F3) were found to be statistically significant. This implies that changes in software configuration and the number of radiologists at both workstations have a substantial impact on system performance.

Moreover, since all three effects are positive, and we are minimizing the objective function, this means that increasing these factor levels generally leads to worse performance. As such, the average main effects are unfavorable in the context of a minimization problem.

4.1.2. Screening Design 2

For the second screening experiment, we built upon the results of the first design by adjusting one of the input factors. In this phase, we compared the current and **new** software configurations at WS5 (Factor 1), while keeping the same variation in the number of radiologists at WS2 (Factor 2) and WS5 (Factor 3).

This screening again employed a 2^3 full-factorial design, systematically testing combinations of high and low values for all three input variables. The eight resulting experimental configurations are shown in *Table 3*.

Design	Factor 1 (Software WS5)	Factor 2 (Radiologists WS2)	Factor 3 (Radiologists WS2)	Response
1	Current (-)	2 (-)	2 (-)	R1
2	New (+)	2 (-)	2 (-)	R2
3	Current (-)	4 (+)	2 (-)	R3
4	New (+)	4 (+)	2 (-)	R4
5	Current (-)	2 (-)	4 (+)	R5
6	New (+)	2(-)	4 (+)	R6
7	Current (-)	4 (+)	4 (+)	R7
8	New (+)	4 (+)	4 (+)	R8

Table 3: Screening design 2: Factor settings and simulation responses

As in the previous screening, simulation results were analyzed using the batch means method to ensure statistical robustness. For each design, we calculated the average objective function, standard deviation, and confidence intervals, and we assessed statistical significance using a t-distribution with 16 degrees of freedom. The full statistical output for this screening design can be found in the Excel sheet **"Screening 2"**.

The results confirm that all three main effects are statistically significant, meaning each factor has a measurable impact on the system's performance. Notably, factor 2 again shows the strongest effect, reinforcing the finding that increasing the number of radiologists at WS2 consistently improves system performance.

As in Screening Design 1, the average effects of all three factors are positive, which is unfavorable in the context of our minimization problem.

4.2. Comparative Design

Following the initial screening experiments, we narrowed our focus to compare specific configurations. The comparison design focuses on evaluating different staffing level and software alternatives against a baseline configuration.

From our screening design analysis, we observed that all main effects showed statistical significance. Thus, comparing different screening designs becomes redundant, suggesting that a mistake may have been made in our analysis. The main effects for all three factors across both screening designs were positive, leading us to conclude that the **Current 2-2** configuration represents the **optimal design choice**. However, for the purposes of continuing this exercise, we have arbitrarily selected the **Upgraded 2-4** configuration as our **baseline**.

We selected this baseline configuration with the following staffing levels (WS2: 2 radiologists, WS5: 4 radiologists) and the upgraded software system. Each alternative configuration was replicated 20 times to ensure statistical validity.

To control the experiment-wise error rate while comparing multiple configurations, we set individual confidence intervals at the level 1 - α/(2×(k-1)), resulting in an individual confidence level of 99.5% for an overall confidence level of at least 90%.
For each comparison, we constructed paired-t confidence intervals using:

$$(X_i - X_s) \pm t_{n-1,1-\frac{\alpha}{2(k-1)}} \times \frac{S_d}{\sqrt{n}}$$

This approach is well-supported in the literature, as Vickerstaff, Omar, and Ambler (2019) note in their BMC Medical Research Methodology paper that when analyzing multiple outcomes in randomized controlled trials, it's important to control the family-wise error rate by adjusting the significance level based on the number of comparisons. Their research confirms that proper control of the experiment-wise error rate is essential when making multiple comparisons to avoid inflating the probability of Type I errors.

The comparative analysis, which is part of the Excel sheet **"Comparative + Selection"**, in our case yielded the following results shown in *Table 4*:

i	Xi - Xs	Halflength	Ci-	Ci+	
Current 2-2	-0,41333	0,04100	-0,45433	-0,37234	Better - Sig
Current 2-4	-0,11214	0,01188	-0,12402	-0,10026	Better - Sig
Current 4-2	0,04111	0,05341	-0,01230	0,09452	Worse – Not Sig
Current 4-4	0,18115	0,02947	0,15168	0,21061	Worse - Sig
Upgrade 2-2	-0,29353	0,02729	-0,32083	-0,26624	Better - Sig
Upgrade 4-2	0,05151	0,03160	0,01991	0,08311	Worse - Sig
Upgrade 4-4	0,29733	0,01831	0,27901	0,31564	Worse - Sig
New 2-2	-0,29660	0,02384	-0,32044	-0,27276	Better - Sig
New 2-4	0,05142	0,01835	0,03307	0,06978	Worse - Sig
New 4-2	0,12833	0,02366	0,10466	0,15199	Worse - Sig
New 4-4	0,34264	0,02891	0,31373	0,37155	Worse - Sig

Table 4: Comparative analysis with 'Upgrade 2-4' as baseline

Based on our comparative analysis using Upgrade 2-4 as the baseline configuration, we identified several configurations that demonstrate statistically significant performance differences. The data reveals that Current 2-2, Current 2-4, Upgrade 2-2 and New 2-2 all showed significant improvement over the baseline. To determine the statistically optimal configuration with a high degree of confidence, we will proceed with these best-performing alternatives in a more rigorous ranking and selection procedure. This will allow us to select the best system with a specified probability of correct selection while accounting for statistical variability in our simulation results.

4.3. Ranking and Selection

Based on the findings from our comparative analysis, we identified several promising configurations for the radiology department. To determine the statistically optimal configuration with a high degree of confidence, we implemented the two-stage ranking and selection procedure developed by Dudewicz and Dalal as explained in *Dudewicz, E. J., & Dalal, S. R. (1975)*.

4.3.1. First stage

This procedure allows us to select the best system from among k alternative configurations with a specified probability of correct selection (P*). For our analysis, we set P* = 0.90, meaning we want to be 90% confident that our selected configuration is truly the best one, provided that the difference between the best and second-best system is at least d* = 0.01 (the indifference amount).

The selection of d* = 0.01 is based on careful consideration of our experimental parameters. Our ranking and selection procedure includes 4 design configurations (k = 4). With k = 4, P* = 0.90, and n0 = 20 replications, the corresponding h-value is 2.583 . According to the ranking and selection methodology, d* must satisfy the relationship: $\mu_2 - \mu_1 \geq d^*$, where μ_1 represents the system with the smallest expected response (average response of current 2-2 configuration) and μ_2 represents the system with the second smallest expected response (average response of upgrade 2-2 configuration). In our case, this difference is approximately 0.12, which is indeed greater than our selected d*.

Furthermore, since we are working with a variance $S^2(20)$ that is less than 0.001 for all five designs, we chose d* = 0.01 to ensure that the Rinott weights W_1 and W_2 remain within the interval [0,1].

Following the methodology, we first conducted n_0 = 20 replications for each of our k = 4 promising configurations. As shown in our analysis, the critical constant h value was determined to be 2.583. For each configuration i, we calculated the first-stage sample means $X_i(20)$ and sample variances $Si^2(20)$. Based on these values, we computed the required total sample size Ni for each system using the formula:

$$N_i = \max\left\{n_0 + 1, \left[\frac{h^2 S^2 i(20)}{(d^*)^2}\right]\right\}$$

This resulted in different required sample sizes for each configuration, ranging from 12 to 39 additional replications. After completing these additional replications, we calculated the weighted sample means $X_i(Ni)$ using appropriate weighting factors, which allowed us to account for the varying precision of our estimates across different sample sizes.

4.3.2. Second stage

To properly account for the varying sample sizes, we computed weighting factors for combining the first and second stage results:

$$Wi1 = \left(\frac{n_0}{Ni}\right)\left\{1 + \left(1 - \frac{n_0}{Ni}\right)\left[1 - \frac{(Ni - n_0)(d*)^2}{h^2 S^2 i(20)}\right]\right\}$$

The final weighted means for each configuration were calculated using:

$$\bar{X}i(Ni) = Wi1 \cdot \bar{X}i(20) + Wi2 \cdot \bar{X}i(Ni - 20)$$

These weighted means provide the optimal combination of the first and second stage information, accounting for the fact that the second stage estimates have different variances.

All of these information can be found in *Table 5*, which can be found in the Excel sheet **“Comparative + Selection”**.

Design	Xi(20)	S^2i(20)	Ni	Xi(Ni-20)	Wi1	Wi2	Xi(Ni)
Current 2-2	-2,44454	0,00051	34	-2,44429	0,60953	0,39047	**-2,44445**
Current 2-4	-2,03879	0,00057	39	-2,03955	0,58811	0,41189	**-2,03911**
Upgrade 2-2	-2,32453	0,00088	59	-2,32915	0,38370	0,61630	**-2,32738**
New 2-2	-2,18975	0,00047	32	-2,18215	0,67514	0,32486	**-2,18728**

Table 5: Two-stage sampling procedure of Dudewicz and Dalal

4.3.3. Select the best system

The configuration with the smallest weighted mean (most negative value) was selected as the best system, with at least 90% probability of correct selection. Based on our results, Current 2-2 with X̄i(Ni) = -2.44445 was determined to be the optimal configuration for the radiology department.

This configuration substantially outperformed the other alternatives, with a weighted mean that was 0.11773 better than the next best configuration (Upgrade 2-2 with X̄i(Ni) = -2.32691). Given that our indifference amount d* was set at 0.01, this difference is not only statistically significant but also practically meaningful.

The two-stage procedure allowed us to make this selection with 90% confidence while efficiently allocating our simulation resources across the different configurations. Configurations with higher variance in their initial estimates (such as Upgrade 2-2) required more additional replications (Ni = 59) to achieve the desired confidence level, while configurations with more stable initial estimates required fewer additional replications (Ni = 32).

This analysis demonstrates the value of structured ranking and selection procedures in simulation studies, as they provide a rigorous statistical framework for selecting the best system when multiple alternatives are being compared.

Since we now know that current 2-2 is the best design, we can make the following conclusion about 4 hypotheses: ${\boldsymbol{H}_0}_1$ and ${\boldsymbol{H}_0}_4$ **can be rejected**, but ${\boldsymbol{H}_0}_2$ and ${\boldsymbol{H}_0}_6$ **cannot be rejected.**

5. Conclusion

This project successfully analyzed and optimized the radiology department using discrete-event simulation. Results confirmed that the **initial design**—especially WS5—was **heavily imbalanced** with long cycle times. Our factorial design revealed that both staffing levels and software at WS5 significantly affect performance. Ultimately, a rigorous ranking and selection procedure demonstrated that the configuration with the **current software, 2 radiologists at WS2, and 2 radiologists at WS5 (Current 2-2) is optimal**. This configuration achieved the best objective function value with at least 90% confidence. Our findings **reject the null hypothesis** ${\boldsymbol{H}_0}_1$ that no better configuration exists and confirm that strategic adjustments can significantly enhance system performance.

6. References

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